WORKED SOLUTION

STEP II 2013, Question 7

Pure • Pell Equation Solutions

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q7

The key idea

Equations of the form $x^2 - 2y^2 = 1$ are Pell equations: once you have one solution, a fixed linear map produces infinitely many more. Parts (i) and (ii) tie these solutions to triangular numbers that are also squares. In part (iii) the trick is to move a^2 across and factorise $c^4 - a^2$ as a difference of two squares. Since b is prime, b^3 can only be split into two factors in very few ways. Part (ii) then provides a non-prime b for the last part.

Solution

Part (i)

The obvious solution is $x = 1$, $y = 0$, since $1^2 - 2 \cdot 0^2 = 1$.

Now suppose $p^2 - 2q^2 = 1$. Then

$$\begin{aligned}(3p + 4q)^2 - 2(2p + 3q)^2 &= (9p^2 + 24pq + 16q^2) - 2(4p^2 + 12pq + 9q^2) \\ &= p^2 - 2q^2 = 1,\end{aligned}$$

since the pq terms cancel. So $x = 3p + 4q$, $y = 2p + 3q$ is also a solution, as required.

Applying this to $(p, q) = (1, 0)$ gives $(3, 2)$, and applying it again to $(3, 2)$ gives $(3 \cdot 3 + 4 \cdot 2,\ 2 \cdot 3 + 3 \cdot 2) = (17, 12)$. Check: $9 - 8 = 1$ and $289 - 288 = 1$. So two solutions with x a positive odd integer and y a positive even integer are

$$(x, y) = (3, 2) \quad \text{and} \quad (x, y) = (17, 12).$$

(In fact the map always keeps x odd, as $3p + 4q \equiv p \pmod 2$, and keeps y even once it is even, as $2p + 3q \equiv q \pmod 2$. So every later solution in the chain also has the right parities: $(99, 70)$, $(577, 408)$, and so on.)

Part (ii)

Let x be odd and y even, and write $x = 2m+1$ and $y = 2n$ with m, n integers. Then $(*)$ becomes

$$(2m+1)^2 - 2(2n)^2 = 1 \iff 4m^2 + 4m = 8n^2 \iff n^2 = \tfrac{1}{2}m(m+1),$$

dividing by 8 in the last step. This is the required form. (Every step is reversible, so the two equations are equivalent for such x and y.)

So n^2 is the triangular number $1 + 2 + \cdots + m$. From part (i), $(3, 2)$ gives $m = 1$, $n = 1$ (so $1 = 1$), and $(17, 12)$ gives $m = 8$, $n = 6$ (so $36 = \frac{1}{2} \cdot 8 \cdot 9$). We use the second of these in part (iii).

Part (iii)

Rearranging and factorising as a difference of two squares,

$$b^3 = c^4 - a^2 = (c^2 - a)(c^2 + a).$$

Since $b^3 > 0$ and $c^2 + a > 0$, the factor $c^2 - a$ is a positive integer too. Also $a > 0$, so $c^2 - a < c^2 + a$.

Ask yourself How many ways are there to write b^3 as a product of two positive integers?

Because b is prime, the only positive divisors of b^3 are 1, b, b^2 and b^3. So the factorisations of b^3 into a smaller factor times a larger one are $1 \times b^3$ and $b \times b^2$. There are no others, and the two factors can never be equal, since b^3 is not a perfect square.

Case 1: $c^2 - a = 1$ and $c^2 + a = b^3$. Adding and subtracting,

$$c^2 = \tfrac{1}{2}(b^3 + 1), \qquad a = \tfrac{1}{2}(b^3 - 1).$$

Case 2: $c^2 - a = b$ and $c^2 + a = b^2$. Adding and subtracting,

$$c^2 = \tfrac{1}{2}b(b+1), \qquad a = \tfrac{1}{2}b(b-1).$$

(The question only asks for these formulae, but it is worth knowing which cases really happen. Case 2 never does. For $b = 2$ it would need $c^2 = 3$. If b is an odd prime, then b and $\frac{1}{2}(b+1)$ are coprime, because $1 \leq \frac{1}{2}(b+1) < b$, and their product is the square c^2, so each must be a square. But a prime is never a square. Case 1 does happen: $b = 23$ gives $\frac{1}{2}(23^3 + 1) = 6084 = 78^2$, so $a = 6083$, $b = 23$, $c = 78$ is a solution.)

The final part

Case 2 fails for primes, but its formulae still give solutions if we drop the requirement that b is prime. Whenever $\frac{1}{2}b(b+1)$ is a perfect square c^2, put $a = \frac{1}{2}b(b-1)$. Then

$$a^2 + b^3 = (c^2 - b)^2 + b^3 = c^4 - 2bc^2 + b^2 + b^3 = c^4 - b^2(b+1) + b^2 + b^3 = c^4,$$

using $2c^2 = b(b+1)$. Part (ii) tells us exactly when $\frac{1}{2}b(b+1)$ is a square: $b = m$ with $n^2 = \frac{1}{2}m(m+1)$. The solution $m = 1$ gives $a = 0$, which is not allowed, but $m = 8$, $n = 6$ works. Take $b = 8$, $c = 6$, $a = \frac{1}{2} \cdot 8 \cdot 7 = 28$:

$$28^2 + 8^3 = 784 + 512 = 1296 = 6^4.$$

So $a = 28$, $b = 8$, $c = 6$ is a solution with $b = 2^3$ not prime.

Checks

- The next solution in the chain, $(99, 70)$, satisfies $9801 - 2 \cdot 4900 = 1$, and it gives $m = 49$, $n = 35$ with $\frac{1}{2} \cdot 49 \cdot 50 = 1225 = 35^2$.
- $6^4 = 1296$ and $28^2 + 8^3 = 784 + 512 = 1296$.
- A computer search over all $c < 400$ found $a = 6083$, $b = 23$, $c = 78$ as the only solution with b prime. It is a Case 1 solution: $78^4 - 6083^2 = (78^2 - 6083)(78^2 + 6083) = 1 \times 12167 = 23^3$.

Common slips

- Forgetting that $c^2 - a$ must be the *smaller* factor, which leads to extra "cases" with negative a.
- Missing the factorisation $1 \times b^3$ and finding only the case $b \times b^2$, or the other way round.
- In the final part, using $m = 1$ from $(3, 2)$, which gives $a = 0$; the question needs a to be positive.

Beyond STEP

The equation $x^2 - 2y^2 = 1$ is Pell's equation. Its solutions are exactly the powers $(3 + 2\sqrt{2})^k = x + y\sqrt{2}$, and the map in part (i) is just multiplication by $3 + 2\sqrt{2}$. This is the start of the theory of units in rings such as $\mathbb{Z}[\sqrt{2}]$ and of continued fractions. Equations like $a^2 + b^3 = c^4$ belong to the study of generalised Fermat equations. For a beautiful introduction to all of this, read Davenport, *The Higher Arithmetic*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.